

AI Affiliate Video Generator

Full Engineering Spec

Personal-use content engine · URL → 3-5 ready-to-upload videos

DOCUMENT	Product Requirements + Engineering Specification
VERSION	v2.1 — Design-Synced Update
DATE	2026-05-02
PRIMARY GOAL	1 Product URL → 3-5 ready-to-upload affiliate videos
NEW IN v2.1	Retry visibility · +5/+10 dual position · Desync UI · Regen Hook spec · Status col

Build the engine first. Ship the content.

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1. Product Definition

The product is a personal-use AI affiliate content machine. It transforms one product URL and optional user-uploaded images into multiple short-form vertical videos, each bundled with ready-to-copy SEO metadata. Version 2 adds five usability improvements without expanding scope beyond the engine.

Core Promise

■ Paste a product URL → click Generate → receive 3–5 videos with title, caption, exactly 3 relevant hashtags, auto-thumbnail, and an MP4 download link. Choose duration, pick an angle, and retry any variant that falls short — all without leaving the results page.

1.1 Target User

Primary user	Solo affiliate creator producing many product videos quickly.
Initial customer	The builder/user personally, before any public SaaS launch.
Usage pattern	Find a product → paste URL → optionally upload images → choose duration + angle → generate variants → review → upload to TikTok manually.

1.2 Product Principles

- **Automation-first, human-polish-last** — the system does 90 % of the work; user only reviews and tweaks.
- **No dependency on product discovery in MVP** — Kalodata-style APIs are future boosters, not core dependencies.
- **Do not build a mini CapCut** — review mode must stay simple, not become a full timeline editor.
- **Dynamic assembly, not slideshow** — every video must include motion, cuts, subtitles, voice-over, and music.
- **Output must be ready to post** — every video includes SEO title, caption, thumbnail, and exactly 3 hashtags.
- **Granular control without complexity** — duration and angle selectors give creative direction without exposing internals.

1.3 Success Definition

Metric	MVP Target
Time from URL to videos	< 5 minutes for 3–5 variants (revised from 2 min; validated at M5)
Video count per generation	3–5 variants by default
Video format	Vertical 9:16 MP4, 1080×1920 preferred
Metadata quality	SEO title + natural caption + exactly 3 product hashtags
Thumbnail	Auto-generated per variant, 1080×1920 JPG
Language support	Indonesian (default) or English, selectable per project
Manual effort	Paste URL → optionally configure → review → copy metadata → download

2. MVP Scope and Non-goals

2.1 In Scope

- Product URL input.
- Optional image upload, max 5 images.
- Basic product extraction: name, images, description.
- AI angle generation with optional user-selected angle type.
- AI script generation: hook, problem, solution, proof/benefit, CTA.
- Video duration selector: Short 15s / Medium 25s / Long 35s.
- TTS voice-over using one default voice per language.
- Dynamic video assembly via FFmpeg with defined motion baseline.
- Auto-subtitles burned in, synced within $\pm 0.5s$ tolerance.
- Thumbnail auto-generated per variant (FFmpeg frame extract).
- Background music from a small curated local library.
- Results page with preview, one-click metadata copy, and MP4 download.
- Retry individual variant without regenerating the full job.
- Generate more variants: +5 and +10.
- Language selector: Indonesian or English.

2.2 Out of Scope for MVP

- AI Product Finder, AI Winning Score, AI Auto Suggest.
- Public billing, subscriptions, credits, or team accounts.
- Full timeline editor or CapCut-like controls.
- Automatic posting to TikTok.
- Custom voice selection (ElevenLabs multi-voice).
- A/B angle performance tracking.
- Large-scale scraping or marketplace crawling.
- Generative AI video from scratch (text-to-video).

2.3 Build Order

Priorit y	Capability	Reason
--------------	------------	--------

P0	URL input + optional images	Entry point for the full workflow.
P0	Video generation pipeline	Core product value.
P0	SEO content package	Makes output ready to post.
P0	Thumbnail generation	Required for manual TikTok upload.
P1	Duration selector	Affects script length and scene count.
P1	Angle type selector	Gives creative direction per variant.
P1	Language selector	Low-effort, high-value for multilingual products.
P1	Review mode	Allows subtitle edits without becoming an editor.
P1	Retry individual variant	Reduces frustration when one of five fails.
P1	Generate more variants	Core personal-use loop.
P2	Data/discovery layer	Future upgrade after engine is validated.

3. User Flow and UX Requirements

3.1 Main Flow

Step 1 — Home / Dashboard: Paste product URL

Step 2 — Configure: Set duration (15s/25s/35s), language (ID/EN), angle type (optional)

Step 3 — Upload (optional): Upload 1–5 product images

Step 4 — Generate: Click Generate — job runs in background

Step 5 — Progress: Status label and progress % shown per stage

Step 6 — Results: 3–5 video cards with preview, thumbnail, metadata

Step 7 — Review: Edit subtitle, regenerate hook, retry one variant

Step 8 — Export: Copy metadata block, download MP4 + thumbnail

3.2 Screen Requirements

Screen	Required Elements	Notes
Dashboard	URL input, duration selector, language selector, angle selector, image upload (with checkmark overlay on uploaded images), Generate button + estimasi waktu helper text, recent projects	Keep clean. Do not overload with analytics.
Generation progress	Status label, progress %, current stage name	Sourced from job_events in queue.
Results grid	3–5 video cards with preview, thumbnail, metadata, copy, download, explicit retry button per card; +5/+10 buttons at top-right AND bottom-center of grid	Each card is self-contained. Retry must be visible directly on card, not hidden in overflow menu.
Review mode — Subtitle tab	Preview, subtitle rows with timestamp + desync warning icon, add/delete row, save & re-render with time warning	No full timeline editor. Red warning icon on rows exceeding ± 0.5 s desync.

Review mode — Regenerate Hook tab	Read-only current hook field, Generate Hook Baru button + helper text, editable new hook result field with Baru badge, Gunakan & Render Ulang button + re-render time warning	Tab must show full content, not be empty.
Projects page	Table with columns: Produk, Durasi, Bahasa, Jumlah Video (with gagal count if incomplete), Status badge, Dibuat, Aksi	Status column: Selesai (green) / Generating (blue) / Gagal (red). Incomplete video count shows warning icon + N gagal.
Generate more	+5 and +10 buttons at top-right and bottom-center of results grid	No credits needed for personal MVP.

3.3 Video Card UI Contract

Each video card must expose the following controls in this order. **Retry must be an explicit visible button on every card** — not hidden inside a three-dot overflow menu.

Element	Behavior
Status badge	Always visible on card. Three states: Siap (green) / Gagal (red) / Sedang Retry (yellow). Must be visible on all cards.
Video preview	Autoplay-muted loop on hover; full audio on click.
Thumbnail preview	Static 1080×1920 JPG shown beside video player.
Title	Editable inline on double-click. Truncated titles show full text on hover (tooltip).
Caption	Editable inline on double-click.
Hashtags (#1 #2 #3)	Read-only display. Exactly 3, always.
[Copy]	Copies title + caption + hashtags as one formatted block.
[Download MP4]	Triggers signed URL download of rendered MP4.
[Download Thumbnail]	Triggers signed URL download of JPG thumbnail.
[Review]	Opens review mode modal for subtitle edit and hook regeneration.
[Retry]	MUST BE EXPLICITLY VISIBLE on card. Re-queues this single variant without touching others. Must be visible on all cards.

■ **Copy behavior (unchanged from v1):** The Copy button copies title, caption, and exactly 3 hashtags as one complete block. It does not copy each field separately. Thumbnail and MP4 have their own separate download buttons.

3.3a NEW — Video Card State Spec

Every video card must handle three distinct render states with clear visual differentiation:

State	Badge	Color	Retry Button	Description
Siap	Siap	Green	Visible, enabled	Variant rendered successfully and ready.
Gagal	Gagal	Red	Visible, enabled	Render failed. User can retry immediately.
Sedang Retry	Sedang Retry	Yellow/Orange	Visible, disabled	Variant is currently being re-queued or re-rendered.

3.3b NEW — Generate More Button Placement

The +5 Video and +10 Video buttons must appear in TWO positions:

- **Top-right of results grid** — visible immediately when results load, before scrolling.
- **Bottom-center of results grid** — visible after user has scrolled through all cards, at the natural end of the list.

3.3c NEW — Regenerate Hook Tab Content Spec

The Regenerate Hook tab in Review Mode must display the following layout in order:

Element	Type	Behavior
Hook Saat Ini label	Muted/gray label	Static text: 'Hook Saat Ini'
Current hook field	Read-only text box	Shows existing hook text. Not editable.
Generate Hook Baru button	Full-width primary purple button	Calls LLM to generate a new hook using same angle. Shows loading state while generating.
Helper text	Muted text below button	'AI akan membuat hook baru berdasarkan angle yang sama'
Hook Baru label	Muted/gray label + Baru badge (green)	Shown only after generation completes.
New hook result field	Editable text box	Shows newly generated hook. User can edit before applying.
Gunakan & Render Ulang button	Primary purple button, right-aligned	Saves new hook and triggers re-render of this variant.
Re-render warning text	Muted helper text below button	'Re-render memakan waktu ~1 menit'

3.4 NEW — Video Duration Selector

A simple three-option toggle exposed on the Dashboard before generation. Selection affects script word count, scene count, and TTS target duration.

Option	Label	Script Target	Scene Count Hint
Short	15s	~40–55 words	4–6 scenes
Medium (default)	25s	~70–90 words	7–10 scenes
Long	35s	~100–120 words	11–14 scenes

3.5 NEW — Angle Type Selector

Optional dropdown shown on the Dashboard. If left blank, the LLM selects angles freely. If selected, all 3–5 generated variants will bias toward the chosen angle type.

Value	Angle Description
(Auto)	LLM selects the best mix of angles for the product.
problem_solution	Open with a pain point, resolve with the product.
testimonial_style	First-person narrative as if from a satisfied user.
comparison	Product vs. common alternative or old method.
how_to	Step-by-step usage demonstration.
benefit_showcase	Lead with top features and end benefits.

4. Functional Requirements

ID	Feature	Requirement
FR-001	Product URL input	User can paste one product URL and submit generation.
FR-002	Optional image upload	User can upload 1–5 images (JPG/PNG/WEBP, max 8 MB each).
FR-003	Basic extraction	System attempts to extract product name, images, and description.
FR-004	Fallback input	If extraction fails after 1 retry, user proceeds with uploaded images + manual product name.
FR-005	Angle generation	System generates 3–5 distinct content angles per product.
FR-006 ◆	Angle type selector	User may optionally select an angle type to bias all variants.
FR-007 ◆	Duration selector	User selects Short / Medium / Long before generation. Affects script and render.
FR-008 ◆	Language selector	User selects Indonesian or English. Affects LLM prompt and TTS voice.
FR-009	Script generation	Each variant has hook, body, and CTA suitable for chosen duration.
FR-010	TTS	Each script produces one voice-over audio file in the selected language.
FR-011	Subtitle generation	Subtitle synced to VO duration within ± 0.5 s tolerance.
FR-012	Video assembly	FFmpeg creates vertical MP4 with defined motion, subtitles, VO, and music.
FR-013 ◆	Thumbnail generation	One 1080×1920 JPG thumbnail auto-generated per variant via FFmpeg frame extract.
FR-014	SEO metadata	Each video gets title, caption, and exactly 3 relevant hashtags.
FR-015	One-click copy	Copy button copies complete metadata block (title + caption + hashtags).
FR-016	Review mode	User can edit subtitles and regenerate hook.

FR-017 ◆	Retry individual variant	User can retry one failed or unsatisfactory variant without affecting others.
FR-018	Generate more	User can request +5 or +10 additional variants.
FR-019	Download MP4	User can download rendered MP4.
FR-020 ◆	Download thumbnail	User can download auto-generated thumbnail JPG.
FR-021	Job status	UI shows generation progress by stage and reports errors per variant.
FR-022 ◇	Video card state badge	Each card displays one of three states: Siap, Gagal, or Sedang Retry with distinct color badges.
FR-023 ◇	Retry button always visible	Retry button must be explicitly visible on each video card — not hidden inside overflow/three-dot menu.
FR-024 ◇	Generate more dual position	+5/+10 Video buttons appear at top-right AND bottom-center of the results grid.
FR-025 ◇	Subtitle desync indicator	Subtitle rows in Review Mode that exceed $\pm 0.5s$ desync tolerance display a red warning icon.
FR-026 ◇	Re-render time warning	Any button that triggers a re-render (Simpan & Render Ulang, Gunakan & Render Ulang) must display muted helper text: 'Re-render memakan waktu ~1 menit'.
FR-027 ◇	Projects status column	Projects page table includes a Status column with badge states: Selesai (green) / Generating (blue-purple) / Gagal (red).
FR-028 ◇	Incomplete variant count	If a project has fewer completed variants than target, show warning icon and 'N gagal' count next to the video count in Projects table.

◆ = new in v2.0 | ◇ = new in v2.1 (from design review)

5. System Architecture

5.1 Recommended MVP Stack

Layer	Choice	Reason
Frontend	Next.js	Fast dashboard, preview UI, API routes.
Backend API	Node.js / Next.js API or Express	Simple integration with AI, TTS, queue, storage.
Queue	BullMQ + Redis	Video rendering must run asynchronously.
Worker	Node.js + FFmpeg child process	Keeps heavy work outside HTTP requests.
Database	PostgreSQL	Reliable relational model for all entities.
Storage	Cloudflare R2 / Supabase / AWS S3	Stores uploads, audio, subtitles, renders, thumbnails.
AI (LLM)	OpenAI GPT-4o or compatible	Angle, script, metadata, validation/fix.
TTS	OpenAI TTS or ElevenLabs	One default voice per language in MVP.
Video	FFmpeg	Low-cost, flexible video assembly.

5.2 Component Diagram

```
[Browser / Next.js UI]
|
v
[Backend API: Node.js]
|-- reads/writes --> [PostgreSQL]
|-- uploads -----> [S3/R2 Storage]
|
v
[BullMQ Queue / Redis]
|
v
[Generation Worker]
|-- calls -----> [LLM API]
|-- calls -----> [TTS API]
|-- runs -----> [FFmpeg]
|-- writes -----> [S3/R2 Storage] (mp4, thumbnail, audio, subtitle)
```

```
|-- updates -----> [PostgreSQL]
|
v
[Results UI: Preview + Thumbnail + Copy + Download]
```

5.3 Deployment Model

- Single app server + one worker process + Redis + PostgreSQL for personal MVP.
- Do not use Kubernetes or microservices for MVP.
- Worker runs on a machine with sufficient CPU and disk IO for FFmpeg.
- Rendering is CPU-heavy; sequential queue (concurrency=1) is sufficient for personal use.
- Split AI/TTS and FFmpeg into separate queues only when render time becomes unacceptable.

6. Sequence Flows

6.1 Generate Initial 3–5 Videos

Step	Actor	Action	Result
1	User	Paste URL, set duration/language/angle, upload images	Frontend validates input.
2	Frontend	POST /api/projects	Project created with config.
3	Frontend	Upload optional images	Asset rows created in storage.
4	Frontend	POST /api/generation-jobs	Job created with target_variant_count, duration, language, angle_type.
5	API	Enqueue BullMQ job	Status = queued.
6	Worker	Extract product data	Product fields and candidate images saved.
7	Worker	Generate angles/scripts/metadata	Variant plans saved with duration and angle constraints.
8	Worker	Generate TTS and subtitles	Audio and subtitle assets saved.
9	Worker	Render videos via FFmpeg	MP4 outputs saved.
10	Worker	Extract thumbnail frame via FFmpeg	JPG thumbnail saved per variant.
11	Worker	Update job complete	UI shows video cards with thumbnail.

6.2 Retry Individual Variant

Step	Action	Result
1	User clicks [Retry] on a single video card	Frontend POST /api/video-variants/{id}/retry
2	API creates a single-variant retry job	Job enqueued, variant status = retrying.
3	Worker re-uses existing angle/script or regenerates if hook retry requested	Audio, subtitle, render, thumbnail re-created.

4	Worker updates variant and file_assets rows	Card in UI refreshes with new output.
---	---	---------------------------------------

6.3 Review and Subtitle Edit

Step	Action	Result
1	User opens review mode	Frontend loads video, subtitle text, metadata.
2	User edits subtitle text	PATCH /api/video-variants/{id}/subtitle saves revised text.
3	User clicks save/render	Single-variant re-render job created.
4	Worker re-renders video + thumbnail	New MP4 and JPG replace previous assets.

6.4 Generate More Variants

Step	Action	Rule
1	User clicks +5 or +10	System creates new job linked to same project.
2	Worker loads existing angles/scripts	New variants must avoid duplicates from avoid-list.
3	LLM generates new angle+hook+script combos	Prompt includes prior angle names and hooks as avoid-list. Duplicate = same angle type AND similar hook (>60% token overlap).
4	Worker renders additional MP4 + thumbnails	Results append to the project grid.

7. Data Model and Database Schema

7.1 Entity Relationship Overview

Each generated video is a variant under a project. Each variant owns a script, metadata, subtitle, media plan, thumbnail, and final MP4 asset.

```
users (optional for personal MVP)
1 -> many projects

projects
1 -> many product_assets
1 -> many generation_jobs
1 -> many video_variants

video_variants
1 -> 1 content_packages
1 -> many file_assets (mp4, audio, subtitle, thumbnail)
1 -> many job_events through generation_jobs
```

7.2 Tables

Table	Purpose	Key Fields / Notes
users	Optional now, required for SaaS	id, email, name, created_at
projects	One product/content project	id, user_id, product_url, product_name, status, language, angle_type, duration_target
product_assets	Source images extracted or uploaded	id, project_id, type, source, storage_key
generation_jobs	Background work record	id, project_id, status, target_variant_count, progress, error
video_variants	One rendered video output	id, project_id, job_id, angle, hook, script_json, subtitle_text, duration_seconds, status
content_packages	SEO metadata per video	id, video_variant_id, title, caption, hashtags (exactly 3 — enforced by DB constraint)

file_assets	Stored files	id, project_id, variant_id, kind (source_image / tts_audio / subtitle / rendered_mp4 / thumbnail), storage_key
job_events	Progress log and debugging	id, job_id, stage, message, metadata, created_at

7.3 Key Schema Changes in v2

The following columns are added to existing tables to support new features:

Table	New Column	Type	Purpose
projects	language	text DEFAULT 'id'	Stores 'id' or 'en' for TTS and LLM language.
projects	angle_type	text NULL	NULL = auto. Stores selected angle type slug.
projects	duration_target	text DEFAULT '25s'	Stores '15s', '25s', or '35s'.
file_assets	kind = 'thumbnail'	(existing enum)	Added 'thumbnail' as valid kind value.
video_variants	retry_count	int DEFAULT 0	Tracks how many times this variant has been retried.

8. API Specification

8.1 Endpoint Summary

Method	Path	Purpose
POST	/api/projects	Create project from product URL with config (language, angle_type, duration_target).
POST	/api/projects/{id}/assets	Upload optional user images.
POST	/api/projects/{id}/generation-jobs	Start generation job.
GET	/api/generation-jobs/{jobId}	Poll job status and progress.
GET	/api/projects/{id}	Get project, assets, variants, metadata.
PATCH	/api/video-variants/{id}/subtitle	Update subtitle text.
POST	/api/video-variants/{id}/regenerate-hook	Regenerate hook, optionally re-render.
POST	/api/video-variants/{id}/retry	NEW — Retry one variant (re-render full pipeline).
POST	/api/projects/{id}/generate-more	Generate +5 or +10 more variants.
GET	/api/file-assets/{id}/download	Signed download URL — MP4 or thumbnail.
GET	/api/health	Basic health check.

8.2 Request / Response Examples

POST /api/projects — updated body:

```
{"product_url": "https://example.com/product/123", "language": "id", "duration_target": "25s", "angle_type": "problem_solution"}
```

POST /api/video-variants/{id}/retry:

```
{"regenerate_script": false} // false = re-render only; true = full re-script + re-render
```

GET /api/projects/{id} — variant shape:

```
{"id": "uuid", "preview_url": "signed-url", "download_url": "signed-url", "thumbnail_url": "signed-url", "title": "SEO title", "caption": "SEO caption", "hashtags": ["#tag1", "#tag2", "#tag3"], "duration_seconds": 24.8, "retry_count": 0}
```

8.3 Validation Rules

- Reject empty URL and invalid URL syntax.
- Block SSRF: reject localhost, private IPs, metadata IPs, non-HTTP protocols.
- Allow image MIME types: image/jpeg, image/png, image/webp only.
- Max 8 MB per image, max 5 images per project.
- Reject duration_target values outside: '15s', '25s', '35s'.
- Reject language values outside: 'id', 'en'.
- Reject angle_type values outside allowed enum or null.
- Reject generate-more count outside allowed list: 5 or 10.
- Never expose raw storage keys to frontend — always return signed URLs.
- Treat product page text as untrusted input against prompt injection.

9. AI, TTS, and Video Generation Pipeline

9.1 Pipeline Stages

Stage	Input	Output	Notes
Extract	Product URL	Product name, image URLs, description	1 retry. Fallback allowed if extraction fails.
Asset normalize	Extracted + uploaded assets	Local normalized images	Resize/crop to 9:16 safe format.
Angle plan	Product data + angle_type	3–5 angles	If angle_type set, bias all angles to type.
Script	Angle + product + duration_target	Script JSON	Word count target per duration tier.
SEO package	Product + script	Title, caption, exactly 3 hashtags	Validate before saving. Regenerate if invalid.
TTS	Script text + language	Audio file	One default voice per language in MVP.
Subtitles	Script + audio duration	SRT/ASS file	Sync within $\pm 0.5s$ tolerance.
Media plan	Assets + script + duration	Scene plan	Scene count from duration tier.
Render	Scene plan + audio + subtitles	MP4	FFmpeg with defined motion baseline.
Thumbnail	Rendered MP4	JPG 1080×1920	FFmpeg frame extract at ~1s mark.
Package	MP4 + thumbnail + metadata	Video card	Preview, Copy, Download, Thumbnail.

9.2 NEW — Thumbnail Generation

- Thumbnail is auto-generated from the rendered MP4 using a single FFmpeg frame extract at the 1-second mark. Output: 1080×1920 JPG stored as kind='thumbnail' in file_assets. No manual step required. Available for download alongside MP4.

FFmpeg command concept:

```
ffmpeg -ss 1 -i render.mp4 -vframes 1 -q:v 2 thumbnail.jpg
```

9.3 NEW — Subtitle Sync Tolerance + UI Indicator

Approximate timing is acceptable in MVP but must stay within a defined tolerance. Maximum allowed desync between subtitle display and TTS audio: ± 0.5 seconds per line. Implementation options (in order of preference):

- **Option A (preferred):** Use OpenAI TTS word-level timestamps if available in API response.
- **Option B:** Estimate timing from character count per scene — $(\text{char_count} / \text{total_chars}) \times \text{audio_duration}$.
- **Option C (fallback):** Even split of audio duration across scenes.

■■ UI Requirement (from design v2.1): In Review Mode, subtitle rows where the calculated desync exceeds ± 0.5 s must display a red warning icon (■) on the right side of the row. This helps users identify and manually fix timing issues without needing to watch the full video. The icon is computed client-side by comparing $(\text{row end_time} - \text{row start_time})$ against TTS segment duration.

9.4 NEW — FFmpeg Motion Baseline

To prevent slideshow feel, all renders must use at least one motion effect per scene. Default baseline filter (zoompan):

```
zoompan=z='min(zoom+0.0015,1.5)':d=90:x='iw/2-(iw/zoom/2)':y='ih/2-(ih/zoom/2)':s=1080x1920
```

Effect	When to Use
zoompan slow zoom-in	Default for product beauty shots.
zoompan slow zoom-out	For reveal or unboxing scenes.
crop pan left-to-right	For wide product images.
crop pan top-to-bottom	For tall product images.

9.5 Video Output Spec

Aspect ratio	9:16 vertical
Resolution	1080×1920 preferred; 720×1280 acceptable for local testing
Duration	15s / 25s / 35s — set by user duration selector
Codec	H.264 video + AAC audio
Container	MP4 with faststart enabled
Frame rate	30 fps

Subtitles	Burned in (ASS format) for TikTok-ready output
Audio	TTS voice-over at normal volume + background music at $\leq -20\text{dB}$ relative
Thumbnail	1080×1920 JPG, extracted at 1s mark

9.6 Script JSON Contract

```
{
  "angle": "problem_solution",
  "target_audience": "pemilik sepatu putih",
  "duration_target": "25s",
  "hook": "Sepatu putih cepat kotor? Ini cara praktis bersihinnya.",
  "scenes": [
    {
      "start_hint": 0,
      "duration_hint": 3,
      "voice_text": "Sepatu putih cepat kotor?",
      "visual_instruction": "Zoom into dirty shoe",
      "subtitle": "Sepatu putih cepat kotor?",
      "motion": "zoompan_in"
    }
  ],
  "cta": "Cek produknya sebelum stok habis."
}
```

10. SEO Content Package Rules

■ ■ HARD RULE: Every video must have one SEO title, one SEO caption, and exactly 3 product-related hashtags. Fewer than 3 or more than 3 is a validation failure.

10.1 Metadata Format

Title
Caption
#hashtag1 #hashtag2 #hashtag3

10.2–10.4 Title, Caption, and Hashtag Rules

Field	Rules
Title	Must contain product category or name. Must include a real benefit. Searchable and not clickbait. Concise and natural in chosen language.
Caption	Must mention the problem and how the product helps. Natural, not robotic, not keyword-stuffed. May include light CTA (e.g. keranjang kuning). Must align with video angle.
Hashtags	Exactly 3. All must relate to category, problem, audience, or use case. No duplicates. No generic spam. All lowercase, no spaces.

10.5 Banned Hashtags

#fyp, #foryou, #viral, #produkviral, #racun, #racunshopee, #racuntiktok, #trending, #viralbanget

10.6 Validation and Fallback Logic

- Validate hashtag count === 3, no duplicates, no banned tags, each is product-related.
- If validation fails: regenerate metadata using stricter prompt (1 retry).
- If still invalid after retry: sanitize and force-select 3 relevant tags from product-specific fallback list derived from product name and category.
- Log validation failures in job_events for observability.

11. Queue, Jobs, and Worker Design

11.1 Job States

State	Meaning	Progress
queued	Job accepted but not started	0–5
extracting	Extracting product data and assets	5–15
planning	Creating content angles and scene plans	15–30
scripting	Generating scripts and SEO metadata	30–45
tts	Generating voice-over audio	45–60
subtitling	Generating subtitle files	60–68
rendering	Rendering MP4 variants	68–92
thumbnailing	Extracting thumbnail frames	92–96
packaging	Creating signed URLs, final metadata	96–99
completed	All target videos completed	100
failed	Job failed after retries	n/a

11.2 NEW — Retry Individual Variant

A retry job targets a single `video_variant` rather than a full project. It re-uses the existing angle and script by default (`regenerate_script=false`) or regenerates from the LLM if requested (`regenerate_script=true`). The retry increments `retry_count` on the variant row and creates a new `job_events` chain. Other variants in the project are unaffected.

11.3 Worker Concurrency

- Personal MVP: start with `concurrency=1` to avoid CPU spikes.
- Scale later by splitting AI/TTS and FFmpeg into separate queues only when needed.
- Thumbnail generation runs in-process after render — no separate queue needed.

11.4 Retry Policy

Stage	Retry Policy	Fallback
Extraction	1 retry	Proceed with uploaded images + manual product name.
LLM	2 retries	Use stricter JSON repair prompt.
TTS	2 retries	Mark variant failed, continue other variants.
FFmpeg render	1 retry	Save render logs, mark variant failed.
Thumbnail extract	1 retry	Skip thumbnail; mark thumbnail_url as null.
Storage upload	2 retries	Keep local temp file until upload succeeds or job fails.

12. Storage and File Lifecycle

12.1 Storage Layout

```
/projects/{project_id}/source/uploaded/{asset_id}.jpg  
/projects/{project_id}/source/extracted/{asset_id}.jpg  
/projects/{project_id}/jobs/{job_id}/plans/{variant_id}.json  
/projects/{project_id}/variants/{variant_id}/voice.mp3  
/projects/{project_id}/variants/{variant_id}/subtitle.ass  
/projects/{project_id}/variants/{variant_id}/render.mp4  
/projects/{project_id}/variants/{variant_id}/thumbnail.jpg ← new in v2
```

12.2 Retention Policy

- Personal MVP: keep all files indefinitely at first if storage is local or cheap.
- Never delete final rendered MP4 or thumbnail until user explicitly deletes project.
- Delete temp directory after successful upload. Keep on failure in dev mode only.
- When SaaS-ready: add retention windows for temporary source files.

13. Security, Validation, and Abuse Prevention

Area	Rule
File upload MIME	Allow only image/jpeg, image/png, image/webp.
File upload size	Max 8 MB per image, max 5 images per project.
File naming	Generate server-side storage keys. Never trust client filenames.
EXIF data	Strip or ignore EXIF metadata during normalization.
URL validation	Validate syntax before extraction. Block localhost, private IPs, metadata IPs, non-HTTP.
Extractor limits	Set request timeout and max response size. Crawl only the single user-provided URL.
Prompt injection	Treat product page text as untrusted. Never allow it to override system rules for hashtag count, banned hashtags, or output schema.
Structured output	Require structured JSON from LLM and validate before saving.
Signed URLs	Never expose raw storage keys. Return signed download URLs only.
Auth (MVP)	Optional local auth. Required for SaaS.

14. Observability and Error Handling

14.1 Metrics to Track

Metric	Why It Matters
Generation duration (URL → all videos ready)	Validates <5 minute target. Replaces over-optimistic 2min target.
Render duration per video	Identifies FFmpeg bottleneck.
Thumbnail extract duration	Should be <1s per variant; flag if slower.
TTS duration	Identifies voice API bottleneck.
Extractor failure rate	Shows URL extraction reliability.
Variant failure rate	Shows render or AI quality problems.
Retry rate per variant	High rate = signal to review angle/script quality.
Manual usable rate	Personal quality metric: how many are good enough to upload.

14.2 Error Categories

Code	Meaning	User-facing Message
EXTRACT_FAILED	Could not extract product data	Could not read product page. Try uploading images or enter product name manually.
AI_SCHEMA_INVALID	LLM returned invalid JSON	Content generation failed. Please retry.
TTS_FAILED	Voice generation failed	Voice generation failed. Please retry.
RENDER_FAILED	FFmpeg failed	Video rendering failed. Please retry or use different images.
THUMBNAIL_FAILED	FFmpeg frame extract failed	Thumbnail unavailable. Video is still downloadable.
UPLOAD_FAILED	Storage upload failed	Could not save generated file. Please retry.

15. Configuration, Folder Structure, and Milestones

15.1 Environment Variables

```
DATABASE_URL=
REDIS_URL=
S3_ENDPOINT=
S3_BUCKET=
S3_ACCESS_KEY_ID=
S3_SECRET_ACCESS_KEY=
OPENAI_API_KEY=
TTS_PROVIDER=
TTS_API_KEY=
APP_BASE_URL=
MAX_UPLOAD_MB=8
MAX_IMAGES_PER_PROJECT=5
DEFAULT_VARIANT_COUNT=5
WORKER_CONCURRENCY=1
DEFAULT_LANGUAGE=id # new
DEFAULT_DURATION_TARGET=25s # new
SUBTITLE_MAX_DESYNC_SECONDS=0.5 # new
THUMBNAIL_EXTRACT_SECOND=1 # new
```

15.2 Folder Structure

```
/apps/web
  /app
  /components
  /lib/api-client.ts

/apps/worker
  /src/index.ts
  /src/jobs/generate-videos.ts
  /src/jobs/retry-variant.ts      ← new
  /src/pipeline/extract-product.ts
  /src/pipeline/generate-angles.ts
  /src/pipeline/generate-scripts.ts
  /src/pipeline/generate-tts.ts
  /src/pipeline/render-ffmpeg.ts
  /src/pipeline/extract-thumbnail.ts ← new
  /src/pipeline/validate-seo-package.ts

/packages/shared
  /src/types.ts
  /src/schemas.ts
  /src/constants.ts
```

```
/infra
  docker-compose.yml
  migrations/
```

15.3 Build Milestones

Milestone	Deliverable	Exit Criteria
M1	Static UI + project creation + config selectors	Can paste URL, select duration/language/angle, create project.
M2	Asset upload + storage	Can upload 1–5 images and preview them.
M3	AI script + SEO package	Can generate validated title, caption, exactly 3 hashtags.
M4	TTS + subtitles (within ± 0.5 s)	Can create voice audio and subtitle file within tolerance.
M5	FFmpeg render with motion baseline	Can render one vertical MP4 with zoompan motion.
M6	Thumbnail generation	Auto-generates JPG thumbnail per variant.
M7	Multi variants	Can generate 3–5 MP4 outputs per job.
M8	Review mode + retry variant + generate more	Can edit subtitle, retry one card, request +5.
M9	Performance validation	Measure end-to-end time. Confirm <5 min for 5 variants.

16. Acceptance Criteria and Test Plan

16.1 MVP Acceptance Criteria

- Given a valid product URL, the system creates a project.
- Given duration_target='15s', scripts target 40–55 words; '25s' = 70–90; '35s' = 100–120.
- Given angle_type='problem_solution', all generated angles bias toward that type.
- Given language='en', TTS produces English audio and LLM generates English scripts.
- A generation job produces at least 3 completed video variants.
- Each MP4 is vertical 9:16 and playable.
- Each video includes voice-over, burned-in subtitles, and motion (no plain slideshow).
- Each video card displays a thumbnail (1080×1920 JPG).
- Each video card has title, caption, and exactly 3 product-related hashtags.
- Subtitle display desyncs by no more than ± 0.5 s from TTS audio.
- Copy button copies one complete metadata block (title + caption + 3 hashtags).
- Thumbnail and MP4 have separate download buttons.
- Hashtag validation blocks or regenerates banned/generic hashtags.
- Retry button on one card re-queues only that variant without affecting others.
- Download button returns the MP4.
- Each video card displays one of three state badges: Siap, Gagal, or Sedang Retry — with distinct colors.
- Retry button is explicitly visible on every video card (not hidden in overflow menu). Disabled state shown during active retry.
- +5 Video and +10 Video buttons appear both at top-right and bottom-center of the results grid.
- In Review Mode Subtitle tab, rows with desync exceeding ± 0.5 s display a red warning icon.
- Any button triggering re-render displays muted helper text 'Re-render memakan waktu ~1 menit'.
- Projects page table includes a Status column with Selesai / Generating / Gagal badge states.
- Projects with incomplete variant count display a warning icon and 'N gagal' count next to the video number.
- Regenerate Hook tab in Review Mode shows: current hook (read-only), Generate Hook Baru button, new hook result field with Baru badge, Gunakan & Render Ulang CTA.

16.2 Manual QA Checklist

Check	Pass Condition
Video feels alive	Contains motion (zoompan/pan), cuts, and subtitles — not static slides.

Hook clarity	First 1–3 seconds communicate product/problem clearly.
Audio mix	Voice is louder than background music.
Subtitle readability	Readable on mobile without zoom. Desync < 0.5s.
Subtitle desync indicator	Red warning icon appears on rows with >0.5s desync in Review Mode.
Thumbnail quality	Clear, well-composed frame — not blurry or mid-transition.
Duration accuracy	Rendered duration within ± 2 s of selected target.
Language consistency	Script, TTS audio, and subtitles all in the same selected language.
SEO title	Includes product category or keyword.
SEO caption	Natural, product-specific, not generic.
Hashtags	Exactly 3, product-relevant, no banned tags.
Copy format	Title, caption, blank line, hashtags — in one block.
Card state badges	Siap / Gagal / Sedang Retry badges clearly visible and color-coded.
Retry visibility	Retry button visible directly on card without opening any menu.
+5/+10 placement	Buttons visible at top-right AND bottom of results grid.
Retry behavior	Other variants unchanged after retry of one.
Projects status col	Status badge updates correctly as job progresses.
Incomplete count	Projects with failed variants show warning + N gagal count.
Regen Hook tab	Full content visible: current hook, generate button, result field, CTA.
Re-render warning	Helper text 'Re-render memakan waktu ~1 menit' visible near every re-render CTA.

16.3 Automated Test Ideas

- Unit test hashtag validator: count, duplicates, banned list, normalization.
- Unit test copy formatter: exact title/caption/hashtag block.
- Unit test duration selector: script word count within target range per tier.
- Unit test subtitle desync: assert ≤ 0.5 s offset per line.
- Unit test subtitle desync indicator: rows exceeding ± 0.5 s return `desync_warning=true`.
- Unit test job state transitions including retry state and Sedang Retry card state.
- Unit test video card state mapping: job status $\rightarrow$ badge state (Siap/Gagal/Sedang Retry).

- Unit test incomplete variant count: project with 3/5 completed returns `gagal_count=2`.
- Integration test: create project → enqueue job → mocked worker → project returns variants with thumbnails.
- Integration test: retry single variant → only that variant updated, others unchanged.
- Integration test: projects list returns correct status badge per project state.
- Smoke test FFmpeg render + thumbnail extract using sample images and sample audio.
- E2E test: Regenerate Hook tab — generate new hook → confirm result field populated → Gunakan & Render Ulang triggers re-render job.

17. Future Roadmap

■ These features must not block MVP. Add them only after the core engine is used personally and proves that the output is worth uploading.

Phase	Feature	Why Later
V2	Template system	After discovering which video structures perform best.
V2	Custom voice selection	After validating default voice quality is the bottleneck.
V2	More review controls	After seeing which edits are most common.
V2	Favorite video → generate similar	Useful once there are winning outputs.
V3	AI Product Finder	Discovery layer after content engine works.
V3	AI Winning Score	Needs real product/video performance signals.
V3	AI Auto Suggest	Retention layer for SaaS version.
V4	Subscription/credits	Only when turning personal tool into SaaS.
V4	Analytics and A/B tracking	Useful after users publish and track results.
V4	Auto-post to TikTok	OAuth complexity not worth it until engine is proven.

Final Build Reminder

■ Build the content engine first. If URL → 3–5 usable videos → copy SEO metadata → download MP4 + thumbnail works smoothly — the MVP is successful. v2.1 closes the gap between PRD and design: card states, retry visibility, dual +5/+10 placement, desync indicator, Regenerate Hook spec, Projects status, and incomplete count warning are all now explicitly documented. Ship the engine. Validate the output. Then grow.